

Solution Architecture

Date	22/09/2026
Team ID	
Project Name	Study Buddy
Maximum Marks	4 Marks

Architecture Layers

Layer	Component	Responsibility
Presentation	React / HTML / CSS / JS	Dashboard and study screens
API	Node.js / Express	REST APIs and business logic
AI	Python AI service / LLM API	Q&A, summaries and quiz generation
Data	MongoDB	Persistent project data
Security	JWT / bcrypt / HTTPS	Secure access and communication

Data Flow

Step	Flow
1	Student → Web UI
2	Web UI → Backend API
3	Backend → Database / AI service
4	AI/Database → Backend
5	Backend → Web UI → Student